Blood Splatter Lab Apparatus

Element A	Information	Other Details	Date Added
A1	Dr Kovacs is a forensics teacher at Richard Montgomery High School. He runs a lab that involves dripping blood onto an angled piece of standard paper to review the splatter. This lab is called the "blood splatter lab". Currently, when doing the lab, students eyeball the angle of the paper, holding it up with their own hands. The current problem is that there is no practical and consistent way to keep the paper at the desired angle. The lab is intended to educate students about the various methods of Forensic Science how one may "recreate the scene" of the murder from blood spatters and the angle at which they were applied. As of now, instead of precisely incrementing their paper to take fairly accurate data, students rely on personal measurements and "eyeballing" the angle at which the paper stands, resulting in somewhat skewed or incoherent data. Thus, our group must design a device that holds a single 8x11-inch piece of paper at a desired angle (incremented by ideally 5 degrees from 0 to 90). The device must be adjustable so that the user can set it to a specific desired angle. The device must also be simple/straightforward for users to access and observe during their lab, as well as allow for paper to be inserted and/or replaced swiftly.	Explanation of how Bloodstain Pattern Analysis works https://science.howstuffworks.com/bloodstain-pattern-analysis3.htm	4/1/25
A2	Dr Kovacs, the High School's lead Forensics teacher, requests a solution to his current unorganized, "lackluster" attempt to alter the angle during the Blood Splatter Lab. Moreover, students struggle to correctly and accurately measure each angle as they drop the blood on their papers. This issue occurs during the student's yearly "Blood Splatter Lab" in Forensic Science Class. Students must release individual blood droplets onto pieces of paper at varying angles, measuring the altered spread of each blood splatter based on these variations. In forensics, professionals analyze the patterns and splatters from real crime scenes to interpret what events took place during the time of the crime, and how the crime was carried out. The angle at which the blood splatter is presented, seen both in the lab and in real-life crime scenes, has the potential to reveal the direction from which the blood came. Furthermore, the angle aids in the almost total reconstruction of the events that caused bloodshed; analysts use the direction and angle of the splatter to establish the "areas of convergence" (the starting point of the bloodshed) and "origin" (the estimation of where the victim and suspect were when bloodshed occurred). Refocusing on the issue at hand, it occurs in his class, as it is the only class that offers this course. This conflict arises due to limited technology/ability to replicate each desired measurement. Students are forced to "eyeball" each angle and often finish the lab with fairly varied information. The issue, although it may seem minute and irrelevant, does have a substantial impact on both the students and teachers participating in the lab, as well as professionals in this field who wish to spread knowledge and interest in such a field, further expanding the scientists in such a field as this field. With an improved "Blood Spatter Set up," students can more accurately gather data, better educating them on the effects and	The impact of altering the angle while dropping blood splatters. https://www.forensicsciencesimplified.org/blood/principles.html#:~:text=By%20measuring%20the%20width%20and,appearance%20of%20the%20resulting%20stain. Lab (which students fill out after observing the blood splatters) https://wranatomyclass.weebly.com/upload	4/1/25

	trends in varying the angle at which blood is dropped upon sheets of paper.	s/1/3/3/7/13374819/lab_2-_blood_spatter_impact_angle.pdf	
A3	The accuracy of forensic blood splatter analysis is crucial in crime scene investigations, as even minor measurement inaccuracies can lead to errors in determining angles of impact and the origin of bloodstains. Traditional methods of manually holding paper at an estimated angle, as the students of Dr. Kovacs' class do, introduce inconsistencies in data collection. According to research on forensic science education, precision in experimental setups directly affects the reliability of student learning outcomes. By constructing a stable, adjustable apparatus, this project addresses the need for improved accuracy and repeatability in forensic science experiments. Studies on experimental forensic methodologies further emphasize the necessity of controlled conditions for results that can be repeated. Using precise angular increments removes most, if not all, of human error, ensuring students' data aligns with the standards of many professional forensic analyses. Works Cited Albright, Thomas, and Charles Gillbert. "How to Make Better Forensic Decisions." *PNAS*, The Rockefeller University, 11 August 2022, https://www.pnas.org/doi/10.1073/pnas.2206567119. Accessed 1 April 2025. Nirupama, Shravani. "Advances in Forensic Bloodstain Pattern Analysis: A Review of Current Methods and Future Directions." Edited by Department of Forensic Science. *INTERNATIONAL JOURNAL OF INNOVATIVE RESEARCH IN TECHNOLOGY*, vol. 11, no. 5, 2024, pp. 2366-2370. https://ijirt.org/publishedpaper/IJIRT168938_PAPER.pdf.	Advances in Forensic Bloodstain Pattern Analysis: A Review of Current Methods and Future Directions https://ijirt.org/publishedpaper/IJIRT168938_PAPER.pdf How to make better forensic decisions https://www.pnas.org/doi/10.1073/pnas.2206567119	4/1/25
A4	(1) The first primary stakeholder would be Dr. Kovacs. Dr. Kovacs, the instructor of the blood splatter analysis lab, introduced the idea to us in the first place. Dr. Kovacs expresses a passion for educating students about forensic science. He feels the need to obtain this device so that students don't need at a minimum of three people to perform this lab. With this device, only one student would be required to do the blood splatter lab. Dr. Kovacs invested in us to find a solution to the lab, so he		4/1/2025

	would be a primary stakeholder as the person. Dr. Kovacs also shares a classroom with another forensics teacher here at Richard Montgomery HS named Mrs. Alba. Mrs. Alba also teaches forensics and her students are participating in the same lab. With this tool, the lab not only helps Dr. Kovacs but also helps Mrs. Alba explain and perform the lab efficiently. (2) The second primary stakeholder would be the students taking Forensic Science and who are participating in this lab. These students would benefit the most from the device, and we need to guarantee that it will work efficiently and that the instructions for using this tool will be very clear. We need to make sure it is durable and works smoothly, so it can work efficiently and consistently. This tool will remove the restrictions set by this lab and will allow an individual student to perform this lab alone, so there's no need to rely on other students to participate in this lab because you can do it alone now. This device will help both Mrs. Alba and Dr. Kovacs' students. (3) The third primary stakeholder is aspiring forensic scientists and people who are really passionate about blood splatter analysis specifically. This device will help educate students and demonstrate a more accurate and realistic blood splatter analysis. This device can be used to compare real blood splatter samples and ones that you generate yourself to allow you to find the most accurate angle. (4) The fourth primary stakeholder would be other forensic science teachers around MCPS. This device can be beneficial for a lot of teachers who want to educate students about this specific blood splatter analysis lab. As Forensic teachers, they all share a common desire to improve the functionality of this lab in particular, as it would significantly relieve the stress and further improperness of the lab as it currently stands.		
A5	Forensic blood spatter analysis plays a critical role in crime scene investigations, influencing legal outcomes and the administration of justice. However, the reliability of this field has been called into question due to issues with inaccurate strategies and insufficient educational resources. The development of an adjustable blood spatter apparatus for forensic science students would therefore directly address these concerns by improving the accuracy of experiments, enhancing forensic education, and ultimately contributing to the integrity of criminal investigations. One of the most significant concerns in forensic science is the potential for false accusations and wrongful convictions due to improper blood spatter interpretation. Studies have shown that errors in bloodstain pattern analysis have led to wrongful enforcement of justice, as subjective and unverified methods have been used in courtrooms instead. The National Institute suggests that “As of 2023, the National Registry of Exonerations has recorded over 3,000 cases of wrongful convictions in the United States.” This number has more than likely grown since the last recording, demonstrating the dire need for improved technologies in order to correctly convict an individual. In many cases, forensic experts have been found to rely on experience-based interpretations rather than controlled, repeatable scientific methods, leading to faulty conclusions. This demonstrates the urgent need for standardized,	https://nij.ojp.gov/topics/articles/impact-false-or-misleading-forensic-evidence-wrongful-convictions https://pubmed.ncbi.nlm.nih.gov/36946413/ https://www.ojp.gov/pdffiles1/nij/grants/228091.pdf https://ejfs.springeropen.com/articles/10.1186/s41935-023	4/2/25

	precise tools that can improve the accuracy of forensic training and subsequent crime scene analysis. Our proposed apparatus will hopefully ensure that students learn to collect data in a controlled environment, reinforcing best practices that will carry over into real-world forensic applications later on in the future. Additionally, the lack of accessible forensic science education has moreover limited opportunities for students to develop strong skills that are seen as being used in many crime scene investigations. Many high school programs struggle to provide hands-on forensic training due to limited access to specialized equipment. This physical interaction is explained to be so vital within Debesh Nilendu's *Egyptian Journal of Forensic Sciences,* in which she states, "Evidence-based education in forensic science acts as a bridge between academia and practice, enhancing training and the practice's quality." Without reliable tools to conduct controlled experiments, students are often left with "theoretical knowledge," lacking practical and physical application of the lessons they are given. The National Research Council mentions in one of their works, *Forensic Science in the United States: A Path Forward,* "They are often strapped in their work, however, for lack of adequate resources, sound policies, and national support. It is clear that change and advancements, both systematic and scientific, are needed in several forensic science disciplines—to ensure the reliability of the disciplines, and establish enforceable standards," With a new, adjustable apparatus, this issue may be addressed by providing an affordable, precise, and easy-to-use solution that enhances forensic science education, making accurate blood spatter analysis more accessible to all. Beyond legal and educational concerns, the scientific credibility of forensic jobs is truly at stake. The National Academy of Sciences, for example, has criticized forensic methodologies, calling for improvements in "data reliability and repeatability." Blood spatter analysis, in particular, has been flagged for lacking thorough "scientific validation." Developing a structured and standardized method for forensic science students to analyze blood spatter under controlled conditions therefore aligns with this recommendation. I will conclude this section with a quote from an article published by NIH, in which the author John Morgan concludes by saying, "In approximately half of the wrongful convictions analyzed, improved technology, testimony standards, or practice standards may have prevented a wrongful conviction at the time of trial." Clearly, there is a need for advanced analysis methods and educational strategies to drastically reduce the number of wrongful convictions. Works Cited "The Impact of False or Misleading Forensic Evidence on Wrongful Convictions." *National Institute of Justice*, 28 November 2023, https://nij.ojp.gov/topics/articles/impact-false-or-misleading-f	-00375-w	

	orensic-evidence-wrongful-convictions. Accessed 2 April 2025. Morgan, John. “Wrongful convictions and claims of false or misleading forensic evidence.” *PubMed*, 22 March 2023, https://pubmed.ncbi.nlm.nih.gov/36946413/. Accessed 2 April 2025. Nilendu, Debesh. “Enhancing forensic education: exploring the importance and implementation of evidence-based education system.” *Egyptian Journal of Forensic Sciences*, vol. 14, no. 6, 2024, https://ejfs.springeropen.com/articles/10.1186/s41935-023-00375-w. “Strengthening Forensic Science in the United States: A Path Forward.” *Office of Justice Programs*, https://www.ojp.gov/pdffiles1/nij/grants/228091.pdf. Accessed 2 April 2025.		
A6	A study by The National Institute suggests that “As of 2023, the National Registry of Exonerations has recorded over 3,000 cases of wrongful convictions in the United States. Organizations such as The Innocence Project work to free the innocent and prevent these convictions, so far exonerating 375 people, including 21 who served on death row.” NIH, an astounding science and medical organization, conducted a different experiment, concluding that “The results are reported of a study to examine case factors associated with 732 wrongful convictions classified by the National Registry of Exonerations as being associated with 'False or Misleading Forensic Evidence.” Furthermore, NIH continues to demonstrate this desire for enhanced technologies/teachings in Forensic Sciences, explaining, “This study, which included the analysis of 1391 forensic examinations, demonstrates that most errors related to forensic evidence are not identification or classification errors by forensic scientists. When such errors are made, they are frequently associated with incompetent or fraudulent examiners, disciplines with an inadequate scientific foundation, or organizational deficiencies in training, management, governance, or resources.” The National Institute of Justice contains a data table that explains the		4/2/25

various methods of Forensic Science, as well as the percentage error found after the study was carried out:

Table 2. The forensic disciplines with the highest percentages of Type 2 errors.

Discipline*	Number of Examinations	Percentage of Examinations Containing At Least One Case Error	Percentage of Examinations Containing Individualization or Classification (Type 2) Errors
Forensic medicine (pediatric sexual abuse)	64	72%	34%
Blood spatter (crime scene)	33	58%	27%
Serology	204	68%	26%
Firearms identification	66	39%	26%
Forensic medicine (pediatric physical abuse)	60	83%	22%
Hair comparison	143	59%	20%
Latent fingerprint	87	46%	18%
Fiber/trace evidence	35	46%	14%
DNA	64	64%	14%